

Use this document to detail specifications/requirements for an item and/or ancillary services. ⓘ

TITLE: 1K helium-4 sorption cooler ⓘ

Requesting Lab/Division/Group: Physical Measurement Laboratory (PML), Applied Physics
Division 686, Faint Photonics Group (Group 6)
NB686080-25-00741

General Statement of Need ⓘ

The Magnetic Imaging Group requires that their two MRI System Phantoms be serviced and upgraded since they have both exceeded their 5 yr useability lifetime. The system phantoms were purchased from CaliberMRI in 2019 for inclusion in the National Medical Phantom Library, hosted by NIST and cosponsored by NIH. The phantoms have been extensively used and they need to be serviced to maintain their desired properties. Also included in the service is an upgrade to allow NIST recalibration using a new protocol that requires embedding recently measured NIST standard samples. The existing phantoms (SN130-0133 and SN130-0134), will be returned to CaliberMRI, the T1, T1, and proton density arrays will be replaced with new solutions. Additional structures will be included to allow NIST to mount standard recalibration vials as required.

NIST develops and maintains medical imaging calibration objects referred to as phantoms. A primary MRI system phantom was developed by NIST and was commercialized. NIST purchased 2 of the commercial MRI system phantoms in 2019 for inclusion into the National Medical Phantom Library jointly sponsored by NIST and NIH. The phantoms have a finite lifetime and require servicing every 5 years.

Items and/or Ancillary Services ⓘ

Item 1: **1K helium-4 sorption cooler**

Quantity: 1

Specifications:

1.1: At least 250uW of cooling at 1K for 24 hours.

1.2: Closed-cycle system that can be used multiple times without adding additional cryogen

1.3: Must have a diameter smaller than 100mm

1.4: Desired Output: Upgraded MRI System phantoms with new reference solutions and new ports for phantom recalibration

1.5: Required Service: Change out of T1, T2, and proton density arrays, addition of new port to allow NIST recalibration of phantoms.

1.6: Performance Standard: Upon delivery to NIST, the phantoms will be imaged and measured to ensure that the service and upgrade were successfully implemented.

Separately Priced Options and Services:

Item 2: [List minimum specifications.

Quantity: [Enter quantity here.]

Specifications: [Enter quantity here.]

Requirements / Specifications: 1K helium-4 sorption cooler

1

Version 1.0 (2019)

Double-click on ⓘ for further assistance.

Click or tap here to enter text.

Service Requirements / Item Specifications

Section ⓘ	Requirements/Specifications ⓘ
Planning Considerations ☒ N/A	☐ List any visits required to allow potential vendors to see site specific constraints. [Explain requirement here] ☐ US citizen required on-site (foreign nationals require preregistration 30 days prior) ☐ Vendor rep on-site more than three days ☐ Other considerations: [List other considerations to ensure an accurate quote]
Shipping & Delivery ☐ N/A	[Enter shipping and delivery information by item here.] Shipping criteria: ☐ Partial delivery acceptable ☐ Direct delivery to building required [Explain need for requirement here] Building has: ☐ Loading dock ☐ Containerization preference: [Choose container] Other requirements: [List any other requirements here] Delivery date criteria: ☒ Delivery of goods, installation, and training shall be completed no later than: 05 Months ARO ☐ Deliver no earlier than: [Insert date here] ☐ Other: [List other shipping, delivery, or specil requirements]
Electronic Media ☒ N/A	[Enter electronic media requirements by item here.] ☐ Software ☐ Electronic manuals ☐ License Key ☐ Other: [List any other Electronic Media requirements here] [Enter email address for software delivery here]
Installation ☒ N/A	[Enter installation requirements by item here.] ☐ Vendor technician on site ☐ Rigging ☐ Uncrating / Unpackaging ☐ Removal of packaging material ☐ Equipment set up ☐ Start-up services ☐ Turn key installation ☐ Other [Enter details here.] Note: When installation is required, TPOC or COR shall be the coordination point.
Facility / Utility Considerations ☒ N/A	[List site specific utility considerations by item here.] ☐ OFPM Work Order Number(s) [List any Work Order Number(s) here]

- ☐ Power available
Voltage: 200 (V)
Max Current: 50 (A)
Phase: Three phase
Frequency: 60 (Hz)
NEMA Plug Type: Unknown (i.e.: NEMA L5-20P)
Environment: Common Lab / Office Environment
Other power consideration: [List any other power considerations here]
- ☐ Utility connections: Cooling water, 6-9L/min @ 5-25C
- ☐ Other considerations: Click or tap here to enter text.

Warranty

☐ N/A

Base Manufacturer Warranty is acceptable

[If you require an Extended Manufacturer Warranty, describe the warranty requirements per item here.]

What is included?

☐ Labor ☐ Parts ☐ Travel ☐ Cost and liability for returns

☐ Other: [Enter other requirements here]

☐ Warranty length: Click or tap here to enter text.

Period Length:[Ex. month, 6 month, year etc.]

Training

☒ N/A

[Training requirements (who, when, what, where) here. If on-site, is travel necessary?]

☐ On-site

☐ Off-site location: [Enter location(s) here]

☐ Number of trainees: [Enter number of trainees here]

Scope of training:

☐ Operations ☐ Maintenance ☐ Troubleshooting ☐ Safety

☐ Other: [Enter other scope requirements here]

☐ Training materials to be provided: [Describe requirement here]

☐ Other: [Enter other training requirements here]

Note: When training is required, TPOC or COR shall be the coordination point.

Inspection and Acceptance

☐ N/A

[Enter inspection requirements per item here]

☐ Planned government acceptance testing (7 business days)

☒ Gov't acceptance expected to take longer than 7 business days: 21 business days

☒ Inspection and acceptance plan: Before inspection can occur, requires installation which may take as much as 2 weeks.

☐ Other: [Enter other requirements here]

Maintenance

☒ N/A

[Describe your maintenance requirements per item here]

Period Length: [Ex. month, 6 month, year etc.]

What is included?

☐ Labor ☐ Spare parts ☐ Consumables ☐ Travel

☐ Cost and liability for returns ☐ Software updates ☐ Manuals	☐ Service plan length: [Enter service plan length here]
☐ Other: [Enter other requirements here]	
☐ Requirements: [Enter requirements here] (i.e.: onsite within 72 hours, etc.)	
☐ Option Period 1 ☐ Option Period 2	
☐ Option Period 3 ☐ Option Period 4	
Government Furnished Material / Equipment ☒ N/A	[List any government furnished material/equipment per item here] ☐ Samples for testing ☐ Government equipment which will leave gov't site (TPOC must coordinate with division property custodian) Serial Number: [Enter Serial Number here] Description: [Enter description here] NIST property number: [Enter NIST property number here] ☐ Other: [Enter other details here]
Travel ☒ N/A	[Describe any travel requirements per item here] ☐ Explain: [Explain here] ☐ Other: [Enter other requirements here] <i>Vendor: To the maximum extent possible, include as part of appropriate sections above (maintenance, repair, etc.). Otherwise, it will be a separate line item, reimbursable per the Federal Travel Regulation.</i>

Provide suggestions to improve this form by clicking [here](#).